Python – Chapter 1 Notes

Python Basics • Beginner-Friendly Study Notes

1. What is Python?

Python is a high-level, interpreted, general-purpose programming language. It is easy to read and is widely used in web development, automation, data science, AI and machine learning.

2. Features of Python

• Easy to learn and read
• High-level language
• Interpreted language
• Dynamically typed
• Object-oriented
• Cross-platform
• Free and open source
• Large library and community support

3. Applications of Python

• Web development
• Data Science
• Artificial Intelligence and Machine Learning
• Automation and scripting
• Testing and DevOps
• Education and research

4. Python Syntax

Python uses indentation to define blocks of code.

```
print("Hello, World!")
```

5. Comments

Comments explain code and are ignored by Python. A single-line comment starts with #.

```
# This is a comment
print("Hello")
```

6. Variables

A variable is a name used to store a value. Python does not require a separate variable-type declaration.

```
name = "Goutam"
age = 19
marks = 85.5
```

7. Rules for Variable Names

• Start with a letter or underscore
• Cannot start with a number
• Can contain letters, numbers and underscores

• Cannot use Python keywords
• Names are case-sensitive

8. Basic Data Types

int → 10   **float** → 10.5   **str** → "Python"
bool → True/False   **list** → [1,2,3]
tuple → (1,2,3)   **dict** → {"name":"Goutam"}
set → {1,2,3}

9. Checking Data Type

The type() function is used to check the data type of a value.

```
x = 25
print(type(x))
```

10. Taking Input

input() takes data from the user. By default, input() returns a string.

```
name = input("Enter your name: ")
print("Hello", name)
```

11. Type Conversion

Type conversion changes one data type into another.

```
age = int(input("Enter age: "))
price = float("99.5")
number = str(100)
```

12. Output using print()

The print() function displays information on the screen.

13. Python Keywords

Keywords are reserved words with special meaning. Examples: if, else, for, while, def, class, return, True, False, None, import and from.

14. Important Exam Points

1. Python is a high-level interpreted language.
2. Python uses indentation for code blocks.
3. Variables do not need explicit type declaration.
4. input() normally returns a string.
5. print() displays output.
6. type() checks the data type.
7. Python is case-sensitive.
8. # starts a single-line comment.

15. Practice Questions

1. What is Python?
2. Write five features of Python.

3. What is a variable?
4. Explain int, float, string and boolean.
5. What is the use of input() and print()?
6. What is type conversion?
7. Why is indentation important?
8. What are Python keywords?
9. Write a program to take name and age as input.
10. Write a program to print Hello World.